Shaan Merchant

(516) 672-5629 merchant.sh@northeastern.edu Shaan Merchant Portfolio

EDUCATION

Northeastern University **Boston, MA**
Bachelor of Science, Computer Engineering (Dean's List & Honors Program) Expected May 2028

- **Related Coursework:** Digital Design & Computer Architecture, Electronics, Embedded Design: Enabling Robotics, Circuits & Signals, Fundamentals of Networks, Program Design & Implementation I & II, Engineering Algorithms

TECHNICAL SKILLS

- **Hardware:** Autodesk Fusion, KiCad, FPGA Development, SystemVerilog, SolidWorks, Oscilloscope, DMM, LTspice, FDM/SLA 3D printing, Waterjet/Routing CNC, NVIDIA Jetson, Arduino, STM32, Soldering, Crimping, Raspberry Pi
- **Software:** Python, C++, Java, TensorFlow, OpenCV, MATLAB, Pandas, Linux, Git, GitHub Copilot, Claude Code

ENGINEERING EXPERIENCE & EXTRACURRICULAR INVOLVEMENT

Iwa Robotics **Los Angeles, CA**
Robotics Engineer Co-op July 2026 - Present

- Designed 3 axis gimbal for camera payloads integrating I2C communication across IMU, encoders, & motor drivers, designing the mount in Autodesk Fusion & tuning PID gains to reduce vibration & jitter
- Developed autonomous navigation & computer vision software on NVIDIA Jetson platforms using agentic AI tools (Claude Code), including GPS/telemetry-based waypoint logging, wild-fire hotspot detection, & cross-drone data handoff for multi-craft survey & response missions
- Built & validated custom payload hardware & PCBs, implementing CAN communication (crimping, connector fabrication, soldering, & circuit debugging) to network motor controllers, sensors, & GPS

Northeastern University Mars Rover Team (NURover) **Boston, MA**
Robotics Engineer Jan 2025 - Present

- Designed power adapter & sensor interface 4-layer PCB in KiCad for robotic arm, optimizing layout for motor controller integration; validated & deployed across 4 of 6 axes in active competition use
- Fabricated auger soil sampling subsystem using SolidWorks, FDM/SLA 3D printing, & CNC routing, integrating temperature/humidity sensor & limit switch with STM32 firmware to meet University Rover Challenge requirements
- Resolved structural failure modes across 3 prototype iterations, validating reliability before deployment in the life detection task & contributing to a 3rd place finish among 110+ international teams

Northeastern University Students for Discovery & Exploration of Space (NUSEDS) **Boston, MA**
Vice President of Projects Sept 2025 - Present

- Coordinated 6 concurrent project teams (NURover, Lunabotics, SUITS, etc.) across a 300+ member organization, tracking progress & providing technical guidance to meet NASA competition deadlines & achieve podium finishes
- Directed budget allocation & alumni relations, partnering with club advisors & Northeastern's College of Engineering to secure funding, mentorship, & career connections that sustain project growth

Northeastern University EECE Department **Boston, MA**
Teaching Assistant - Embedded Design: Enabling Robotics (EECE 2160) Sept - Dec 2025

- Led instruction & mentorship for 90+ students by teaching Field-Programmable Gate Array (FPGA) fundamentals, including logic gates & digital circuit design, strengthening students' hardware design proficiency
- Directed development of six-legged FPGA-controlled robots with 18 servo motors in Embedded C++, authoring 10 original lab assignments covering PWM motor control & sensor integration, adopted into the course curriculum

PROJECTS

Hexapod Robot Control System | DE1-SoC FPGA, C++, Embedded Linux, May - June 2025

- Implemented memory-mapped FPGA servo control circuit driving 18 servo motors, with object-oriented C++ embedded control software on Linux executing forward, backward, left, right, & initialization movements
- Integrated obstacle avoidance sensor for basic perception mapping & tested multiple servo positions to tune optimal movement profiles for reliable motion

Plant Guardian, Agricultural Disease Detection System | Python, TensorFlow, Jetson Nano Sept 2023 - Aug 2024

- Trained TensorFlow CNN on 2,600+ images to classify 5 plant diseases with 99.5% accuracy, deploying on NVIDIA Jetson Nano with TensorFlow Lite & YOLOv8 for real-time field inference
- Built battery-powered detection device with camera, GPS, & SMS modules that identified diseases in field, texting farmers the crop location & treatment options